
Lifted Cavity Networks: In-Context Sensor Fusion with Missing Sensors and Shared Noise

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Low-cost sensor deployments must estimate a target from whichever sensors are
2 working. They have only a short calibration record, and temperature, humidity
3 and drift perturb several sensors at once. We cast this as in-context probabilistic
4 prediction, and ask which architectures can be trained once across deployments yet
5 remain correct for every missingness pattern and any number of sensors. Products
6 of per-sensor experts over the target meet both requirements, but we show they
7 cannot be Bayes-consistent across missingness patterns when sensors share noise,
8 and an expectation-propagation network with scalar cavities fails accordingly in
9 practice. We propose *lifted cavity networks*: each observed sensor contributes
10 a rank-one Gaussian site over the target and a low-dimensional shared nuisance,
11 which is exact for factor-structured noise under every pattern. The sites are ini-
12 tialized from a supervised factor analysis of the incomplete calibration rows and
13 refined by a learned, cavity-conditioned re-linearization. On synthetic tasks with
14 exact and Bayes-optimal references, an 8k-parameter lifted cavity network closes
15 58% of the gap between the closed form and the Bayes-optimal predictive when two
16 sensors are missing, a pattern never seen in training. A 25-times-larger TabPFN-
17 v2-style transformer trained on the same task distribution closes 34%. The lifted
18 network also transfers from five sensors to sixteen. On two real sensor networks it
19 beats every closed form after fine-tuning, but a fine-tuned transformer is better still.
20 We therefore test lifted sites as the output layer of the transformer; that evaluation
21 is in progress. Both evaluation protocols were committed before their test data
22 existed.

1 Introduction

24 Low-cost sensors now carry much of the world’s environmental, industrial and clinical measurement.
25 A typical deployment co-locates a few cheap sensors with a reference instrument for a short calibration
26 period, and from then on predicts the reference quantity from the sensors alone. Three features
27 make this hard. First, every deployment is different: gains, cross-sensitivities and noise depend
28 on the unit, its placement and its age, and the calibration record is short. Second, sensor noise is
29 *shared*: temperature, humidity, local emissions and drift move several sensors at once, so sensors are
30 not conditionally independent given the target. Third, sensors fail, are serviced or are added, so at
31 prediction time an arbitrary subset is available, often one never seen together during calibration.

32 We study this as *in-context* probabilistic prediction. A network receives a deployment’s labeled
33 calibration rows (the support), with whatever entries are missing, and a query made of an arbitrary
34 subset of sensor readings, and it returns a predictive distribution for the target. Trained once across
35 many deployments, such a network adapts to a new one from its calibration rows alone, as conditional
36 neural processes and prior-fitted networks do for regression and tabular prediction [8, 26, 15].

37 The problem imposes two structural requirements. The predictor must be *mask-consistent*: defined
 38 and accurate for every subset of observed sensors, including subsets absent from training. It must also
 39 be *width-agnostic*: usable for any number of sensors. A product of per-sensor experts over the target
 40 meets both by construction, because a missing sensor simply contributes no factor [13, 34]. Learned
 41 message passing in the style of expectation propagation (EP) refines such experts using context from
 42 the other sensors [24, 12].

43 Contributions.

- 44 1. **When products of experts fail (§3–4).** Under shared noise, no product of mask-independent
 45 experts over the target alone is Bayes-consistent across missingness patterns; the optimal
 46 weight of a sensor can even change sign with the other sensors present. More generally,
 47 Gaussian sites over the target and K' nuisance dimensions are consistent only if K' reaches
 48 the rank of the shared noise, and then they are exact. An EP-style network with scalar
 49 cavities fails accordingly in practice.
- 50 2. **Lifted cavity networks (§4).** Each observed sensor contributes a rank-one Gaussian site
 51 over the target and a low-dimensional shared nuisance, and missing sensors are omitted
 52 exactly. The sites start from a factor analysis of the incomplete calibration rows and are
 53 refined by a small network that re-linearizes each sensor’s response under its cavity.
- 54 3. **A pre-registered evaluation (§6)** against exact posteriors, a Hamiltonian Monte Carlo esti-
 55 mate of the Bayes-optimal predictive, tuned closed forms and a TabPFN-v2-style transformer
 56 trained on the same task distribution, on synthetic panels and two real sensor networks.
 57 Three protocols were committed before their test data existed, and every endpoint is reported.

58 **What we find.** Structure buys sample efficiency and extrapolation, but on real data it costs flexibility.

- 59 • On synthetic tasks, the 8k-parameter lifted cavity network beats a 203k-parameter TabPFN-
 60 v2-style transformer trained on the same task distribution for every number of missing
 61 sensors. It closes 58% of the achievable gap against 34%, and the margin grows when it
 62 runs on 16 sensors after training on five.
- 63 • The advantage vanishes under strong response nonlinearity and when sensors are linear
 64 (§6.3).
- 65 • On two real sensor networks, the fine-tuned lifted network beats every closed form, but the
 66 fine-tuned transformer is better, and one of our pre-registered endpoints fails (§6.5).

67 The last finding motivates a third protocol (§6.6), which uses lifted sites as the output layer of the
 68 transformer.

69 2 Problem setup

70 A task (deployment) $\tau \sim p(\tau)$ defines a joint distribution $p_\tau(x, y)$ over P sensor readings $x \in \mathbb{R}^P$
 71 and a scalar target y . The support $\mathcal{D} = \{(x_i, m_i, y_i)\}_{i=1}^n$ holds n labeled rows whose entry x_{ij} is
 72 observed where $m_{ij} = 1$. A query is a reading x_S of an observed subset $S \subseteq \{1, \dots, P\}$. The target
 73 of prediction is the posterior predictive

$$p(y \mid x_S, \mathcal{D}) = \int p_\tau(y \mid x_S) p(\tau \mid \mathcal{D}, x_S) d\tau, \quad (1)$$

74 and methods are scored by the negative log-likelihood (NLL) of the held-out target. Two privileged
 75 references bound what is possible on synthetic tasks: the *oracle* $p_\tau(y \mid x_S)$, which knows the task,
 76 and the *Bayes-optimal predictive* (1), which knows only the task prior. We call the NLL difference
 77 between the closed-form FA-Gaussian of §4 and the Bayes-optimal predictive the *achievable gap*.

78 An in-context method learns a map $(\mathcal{D}, x_S) \mapsto q(y)$ by minimizing NLL over tasks drawn from a
 79 training distribution [8, 26, 9]. Throughout, models are trained only on queries with *at most one*
 80 missing sensor, and evaluated on patterns with up to three missing. Two- and three-sensor deletions
 81 are therefore never seen in training.

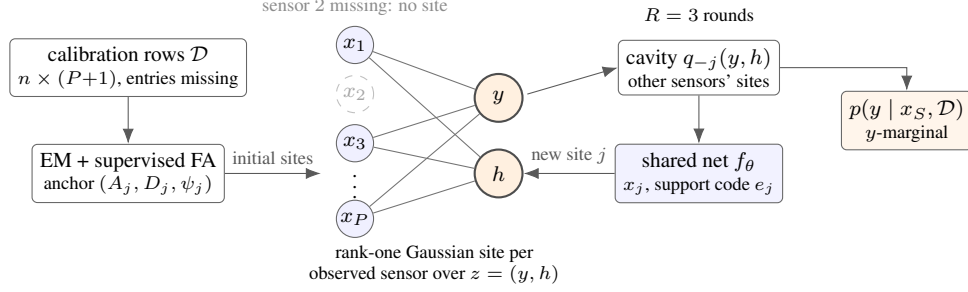


Figure 1: Lifted cavity network. A supervised factor analysis estimated from the incomplete calibration rows gives every sensor a rank-one Gaussian site over the target y and a shared nuisance h . Missing sensors contribute no site, so any subset and any number of sensors are handled by omission. For R rounds, a shared network replaces each observed sensor’s site using its cavity (the posterior from the other sensors) and an encoding of that sensor’s calibration scatter. The prediction is the y -marginal of the product.

3 Scalar experts cannot fuse sensors with shared noise

Consider the linear-Gaussian model $x_j = a_j y + d_j u + \sigma_j \epsilon_j$ with $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$, where u is a nuisance shared by all sensors. A product-of-experts (PoE) predictor gives each sensor a factor $t_j(y; x_j)$ and predicts $q(y | x_S) \propto p(y) \prod_{j \in S} t_j(y; x_j)$. Omitting missing sensors is what makes the construction mask-consistent.

Proposition 1 (Scalar experts). *Let each t_j be Gaussian in y with positive precision, depend on x_j , and not depend on S . Suppose q equals the Bayes posterior for every single-sensor query $S = \{j\}$. Then for a pair $\{j, k\}$ with $a_j a_k \neq 0$, q equals the Bayes posterior if and only if $d_j d_k = 0$. Moreover, q ’s pair posterior mean gives x_j a coefficient with the sign of a_j , whereas the Bayes coefficient can have the opposite sign.*

The proof (Appendix A) shows that matching single-sensor queries forces $t_j(y; x_j) \propto p(x_j | y)$, so q is the naive-Bayes posterior. For example, take $a = (1, 1)$, $d = (1, 0.5)$ and $\sigma = (0.05, 0.05)$. The Bayes coefficient of x_1 is $+0.50$ when x_1 is alone and -0.96 in the pair: the second sensor lets the predictor estimate u and subtract it from the first.

EP [24] refines each factor using its *cavity*, the approximate posterior given the other sensors, and learned EP amortizes the refinement [12, 6, 16]. With a Gaussian over y alone, however, the cavity summarizes the other sensors by a scalar mean and variance, discarding what they reveal about u . Proposition 1 does not exclude every scalar-cavity network, but the failure is visible in practice (§6.4).

4 Lifted cavity networks

Lifted sites. Let $z = (y, h) \in \mathbb{R}^{1+K}$, where h is a K -dimensional nuisance with prior $\mathcal{N}(0, I)$ and y has prior $\mathcal{N}(\mu_y, v_y)$. Write the prior in natural parameters (Λ_0, η_0) . Each observed sensor contributes a rank-one Gaussian site, which corresponds to a linear-Gaussian measurement $\tilde{x}_j = w_j^\top z + e_j$ with $e_j \sim \mathcal{N}(0, \psi_j)$:

$$\Lambda_j = \frac{w_j w_j^\top}{\psi_j}, \quad \eta_j = \frac{w_j o_j}{\psi_j}. \quad (2)$$

The prediction is the y -marginal of

$$q(z | x_S) = \mathcal{N}(z; \Lambda^{-1} \eta, \Lambda^{-1}), \quad \Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j, \quad \eta = \eta_0 + \sum_{j \in S} \eta_j. \quad (3)$$

Proposition 2 (Exactness). *Suppose $x_j = A_j y + D_j^\top h + c_j + e_j$, with independent $e_j \sim \mathcal{N}(0, \psi_j)$ and the priors above. Set $w_j = (A_j, D_j)$ and $o_j = x_j - c_j$. Then (3) equals $p(z | x_S)$ for every $S \subseteq \{1, \dots, P\}$, and its y -marginal is the Bayes predictive for every missingness pattern.*

Shared noise of rank K is absorbed by K nuisance dimensions, so the obstruction of Proposition 1 disappears: the lifted product differences out h automatically. Fewer nuisance dimensions cannot suffice, so the lifting dimension must match the rank of the shared noise:

Proposition 3 (Lifting dimension). *Let $x = ay + Du + e$ with $D \in \mathbb{R}^{P \times K}$, $u \sim \mathcal{N}(0, I_K)$, independent Gaussian e , and every $a_j \neq 0$. Suppose that after deleting any row, $L = [a, D]$ still contains two disjoint nonsingular $(1 + K) \times (1 + K)$ submatrices, which holds for almost every L once $P \geq 2K + 3$. If a product of mask-independent sites of the form (2) with $o_j = x_j - c_j$, over $z = (y, h)$ with $h \in \mathbb{R}^{K'}$ and a Gaussian prior, reproduces $p(y \mid x_S)$ for every S with $|S| \leq 2$, then $K' \geq K$.*

The proof (Appendix A) shows that the conditionals on single sensors and pairs identify the covariance of the sensors, which a product of such sites can only represent as a matrix of rank $1 + K'$ plus a diagonal; the identifiability argument of factor analysis [1] excludes this when $K' < K$. Our main synthetic panels ($K = 1$, $P = 5$) meet the condition, so for products of static sites one nuisance dimension is necessary and sufficient.

Closed-form initialization from the support. Before any learning, we estimate the factor model from $\mathcal{D}$ alone. EM for a Gaussian with missing entries [4, 21] gives the joint mean and covariance of (x, y) from the incomplete calibration rows. Then $A = \text{Cov}(x, y) / \text{Var}(y)$, and principal-factor iterations on the residual covariance $\text{Cov}(x) - AA^\top \text{Var}(y)$ give (D, ψ) [31]. With sensors standardized by their support statistics, $o_j = \tilde{x}_j + A_j \mu_y$. We call the resulting predictor *FA-Gaussian*; it is exact for every mask under the fitted model.

Learned cavity refinement. Real sensors are nonlinear, real noise is not exactly factor-structured, and $n = 48$ calibration rows make plug-in estimates noisy. We therefore refine the sites for R rounds. In each round, every observed sensor j removes its site from (3) to form the cavity $q_{-j}(z) = \mathcal{N}(m_{-j}, V_{-j})$, the posterior over target and nuisance given the other sensors. The cavity predicts the reading as $\hat{x}_j = A_j(m_{-j,y} - \mu_y) + D_j^\top m_{-j,h}$, with variance $v_j = w_j^\top V_{-j} w_j + \psi_j$. A shared two-layer network f_θ reads the reading $\hat{x}_j$, an encoding e_j of sensor j 's calibration scatter, the anchor $(A_j, D_j, \log \psi_j)$, the cavity $(m_{-j}, \log V_{-j,yy}, V_{-j,yh})$, the raw and standardized residuals of $\tilde{x}_j$ against $\hat{x}_j$, $\log v_j$ and the fraction of observed sensors. It outputs a correction $(\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j)$, and the site is replaced by the local linearization of the sensor's response at the cavity mean $\bar{y} = m_{-j,y}$:

$$\begin{aligned} w_j &= (A_j + \delta_j^{\text{slope}}, D_j + \delta_j^D), & o_j &= \tilde{x}_j - g_j(\bar{y}) + w_{j,y} \bar{y}, \\ g_j(\bar{y}) &= A_j(\bar{y} - \mu_y) + \delta_j^{\text{off}}, & \psi_j &\leftarrow \psi_j e^{4 \tanh(s_j/4)}. \end{aligned} \quad (4)$$

The encoding $e_j \in \mathbb{R}^{16}$ is a DeepSets network [35] over sensor j 's calibration scatter, so f_θ can recognize a sensor's nonlinearity from the support. The last layer of f_θ is initialized at zero.

If f_θ were replaced by the statistical linear regression of a *known* response function under the cavity, these rounds would be EP with posterior linearization [3, 7]. The network amortizes the unknown response from the calibration rows, in the spirit of algorithm unrolling [11, 25].

Proposition 4 (Structural guarantees). *For any parameters θ :*

- (i) *the prediction does not depend on the readings of missing sensors;*
- (ii) *permuting sensors together with their calibration columns leaves the prediction unchanged, and the same θ applies for every P ;*
- (iii) *when the last layer of f_θ is zero, the network equals the FA-Gaussian closed form of its anchor, so the minimal training risk over θ is at most that closed form's.*

Training. We minimize the predictive NLL over episodes drawn from a pool of 60,000 synthetic tasks (128 tasks $\times$ 8 queries per step, 20,000 AdamW steps with cosine decay, query masks with at most one missing sensor). We use $K = 1$ and $R = 3$. The model has about 8,000 parameters and trains in 15–20 minutes on one CPU core, 14 times less than the transformer baseline (Table 6). The same recipe trains every learned baseline except the transformer, which gets fresh tasks at every step.

5 Related work

Products of experts handle missing modalities by omission [13, 34, 22]; our negative result says when experts over the target alone are insufficient. **Learned message passing**, through learned EP messages [12, 6, 16] or unrolled inference [11, 25], amortizes inference inside a fixed model; here the model is unknown and must be read from a small calibration set, and the cavity is over a lifted latent. **Supervised learning with missing values** [30, 21, 17] studies one task at a time; NeuMiss [20] unrolls the Bayes predictor of a Gaussian model but is trained per dataset and keyed to feature identities. **In-context prediction** with neural processes [8, 19] and prior-fitted networks such as TabPFN and TabICL [26, 14, 15, 28] uses generic set or transformer architectures; we compare against such a transformer trained on our task distribution and test lifted sites as its output layer. **Fusion under unknown correlations**: covariance intersection [18] bounds fused errors, and learned fusion with correlated noise [33, 5] or learned Kalman gains [29] train per system with fixed sensor sets.

6 Experiments

6.1 Setup

Synthetic tasks. Each task draws, for every sensor j , a response $x_j = a_j y + b_j + 0.4 \sin(f_j y) + d_j u + \sigma_j \epsilon_j$ with $a_j = \pm \mathcal{U}[0.6, 1.8]$ (random sign), $b_j \sim \mathcal{N}(0, 0.4^2)$, $f_j \sim \mathcal{U}[0.5, 2.5]$, shared-noise loading $d_j \sim \mathcal{N}(0, 0.7^2)$, $\sigma_j \sim \mathcal{U}[0.15, 0.8]$ and $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$. The support has $n = 48$ rows with 20% of sensor entries missing at random, and 48 query rows are scored under every mask with k missing sensors (or 20 seeded masks when there are more). The main panel (F1) has 256 fresh tasks with five sensors. Six further panels of 128 tasks change one factor at a time: 8 or 16 sensors, nonlinearity 0.8 or 0, and 16 or 96 support rows.

Methods. *Support-only closed forms*: FA-Gaussian, EM-Gaussian, NL-FA (cubic responses with a tuned curvature penalty and exact grid inference, a strong plug-in nonlinear model), evidence-tuned Bayesian linear regression (BLR), ridge, and, post hoc, a Gaussian process. *Learned in-context models*, all trained on the same task family and masks: our lifted cavity network (8k parameters, three seeds); a residual MLP that receives the same FA anchor and support encodings (15k parameters, five sensors only); the scalar-cavity network of §3; and a TabPFN-v2-style cell-token transformer [15] trained on fresh tasks at every step (203k parameters, 30,000 steps). A second transformer was trained on *all* masks, to separate architecture from training coverage. *Privileged references*: the oracle, and the Bayes-optimal predictive estimated by HMC with the shared nuisance marginalized analytically (Appendix C).

Protocol. NLL is the primary metric; coverage, CRPS [10] and RMSE are secondary. Intervals are paired 95% intervals over tasks, with per-task NLL averaged over training seeds. Three evaluation protocols were committed to version control before their panels existed, and every endpoint is reported (Table 5, Appendix D).

6.2 Synthetic tasks with exact references

Table 1 and Figure 2a report the main panel. The lifted cavity network is the best non-privileged method for every number of missing sensors, including the two- and three-sensor deletions it never saw in training. With two missing sensors it closes 58% of the achievable gap between FA-Gaussian and the Bayes-optimal predictive. Two comparators close less: the residual MLP, which receives the same anchor and support encodings with twice the parameters (46%), and the tuned nonlinear plug-in model NL-FA (19%). The TabPFN-v2-style transformer has 25 times the parameters and sees fresh tasks at every step. It is still worse than the lifted cavity network at every number of missing sensors, by 0.020 to 0.033 nats (all intervals exclude zero), and it closes 34% of the gap. Training it on every mask pattern does not change this (33%): its deficit comes from how efficiently it learns fusion from a finite training budget, not from unseen patterns. The residual MLP is the closest competitor in this setting. The lifted cavity network beats it by 0.030 nats with all sensors observed and by 0.016 with two missing, and the two tie with three missing. EM-Gaussian, BLR and both ridges fall below

Table 1: Main synthetic panel (F1: 256 fresh tasks, five sensors). Mean test NLL by number of missing query sensors; learned models saw at most one missing sensor in training, so $k = 2, 3$ are unseen patterns. Learned rows average three training seeds (seed s.d. ≤ 0.002) except the transformers (one seed). Last column: share of the achievable gap (FA-Gaussian to Bayes-optimal) closed at $k = 2$. Best non-privileged entry per column in bold.

Method	$k=0$	$k=1$	$k=2$	$k=3$	Gap closed
<i>Privileged references</i>					
Oracle (true parameters)	-0.419	-0.250	-0.026	0.287	-
Bayes-optimal (HMC, true prior)	-0.334	-0.174	0.039	0.337	-
<i>Support-only closed forms</i>					
FA-Gaussian	-0.152	-0.011	0.177	0.439	0%
EM-Gaussian	-0.048	0.037	0.195	0.441	-13%
NL-FA (cubic, tuned)	-0.173	-0.036	0.151	0.417	19%
Bayesian linear regression	0.111	0.091	0.214	0.450	-26%
Ridge (complete case)	0.026	0.078	0.216	0.451	-28%
Ridge (mean-imputed)	0.198	0.280	0.399	0.582	-161%
GP, complete case [†]	-	-	-	-	-
<i>Learned in context (trained on $k \leq 1$)</i>					
Lifted cavity (ours), 8k	-0.265	-0.112	0.097	0.395	58%
Transformer (TabPFN-v2 style), 203k	-0.246	-0.084	0.131	0.424	34%
Residual MLP on FA anchor, 15k	-0.236	-0.087	0.114	0.397	46%
Scalar-cavity network, 3k	0.055	0.178	0.336	0.555	-115%
<i>Transformer, trained on all masks</i>	-0.237	-0.076	0.132	0.411	33%

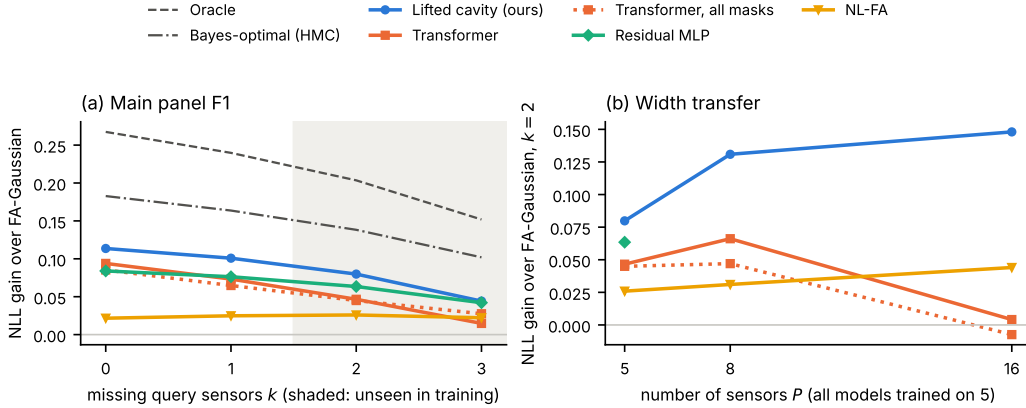


Figure 2: NLL gain over the closed-form FA-Gaussian anchor (higher is better). (a) Main panel F1 by number of missing query sensors; shaded patterns were never seen in training. Grey lines are the privileged oracle and the Bayes-optimal predictive. (b) Width transfer with two missing sensors: every learned model was trained on five sensors, and the residual MLP exists only at $P = 5$.

204 FA-Gaussian, because they cannot share strength across sensors under missingness. Mean imputation
205 of calibration entries is the worst of all.

206 6.3 Generalization beyond the training distribution

207 Every learned model was trained on five sensors, nonlinearity 0.4 and 48 calibration rows (Table 2,
208 Figure 2b).

209 **Width.** The lifted cavity network runs unchanged on 8 and 16 sensors, where it improves on FA-
210 Gaussian by +0.131 [+0.097, +0.165] and +0.148 [+0.115, +0.181] nats with two missing sensors.
211 The residual MLP cannot run on these panels, and EM-Gaussian degrades as the covariance grows.
212 The transformer also runs at any width, but it gains less over FA-Gaussian as sensors are added.

Table 2: Generalization beyond the training distribution (fresh panels of 128 tasks). Mean test NLL with two missing query sensors. Every learned model was trained on $P=5$ sensors, nonlinearity 0.4 and $n=48$ support rows; the residual MLP is defined only for five sensors. Best non-privileged entry per column in bold.

Method	$P=8$	$P=16$	nonlin. 0.8	linear	$n=16$	$n=96$
Oracle	-0.598	-1.142	-0.086	-0.066	-0.060	-0.043
FA-Gaussian	-0.296	-0.797	0.334	0.009	0.510	0.114
EM-Gaussian	-0.160	2.977	0.311	0.034	1.140	0.112
NL-FA	-0.327	-0.841	0.258	0.030	0.496	0.073
BLR	0.615	0.058	0.331	0.061	3.690	0.125
Lifted cavity (ours)	-0.427	-0.945	0.191	0.031	0.363	0.052
Transformer	-0.362	-0.801	0.191	0.085	0.523	0.081
Residual MLP	-	-	0.235	0.036	0.371	0.074
<i>Transformer, all masks</i>	-0.343	-0.790	0.190	0.080	0.483	0.080

Table 3: Real sensor networks: mean test NLL on later, held-out periods. Beijing: 12 target stations $\times$ weekly episodes in 2016–2017 with natural missingness, plus extra dropped sensors. Air Quality: weekly episodes after October 2004, sensor dropout simulated. Fine-tuning uses only earlier periods and one recipe for every model; lifted-cavity and residual-MLP rows average three seeds. Best entry per column in bold.

Method	Beijing PM _{2.5} (11 stations)			Air Quality CO		Air Quality NO ₂	
	natural	+3	+6	$k=0$	$k=2$	$k=0$	$k=2$
EM-Gaussian	1.032	0.742	0.569	0.074	0.085	-0.237	-0.175
FA-Gaussian	0.677	0.615	0.562	0.052	0.116	-0.196	-0.155
NL-FA	0.938	0.744	0.610	0.144	0.091	-0.090	-0.142
Bayesian linear regression	0.399	0.406	0.442	1.037	0.216	-0.055	-0.178
Ridge (complete case)	0.425	0.432	0.458	0.824	0.208	-0.200	-0.176
GP, complete case [†]	-	-	-	3.364	0.503	-	-
Lifted cavity, zero-shot	0.826	0.661	0.561	0.044	0.129	-0.350	-0.233
Transformer, zero-shot	0.958	0.743	0.687	-0.017	0.081	-0.360	-0.231
Lifted cavity, fine-tuned (ours)	0.205	0.237	0.305	-0.040	0.008	-0.412	-0.337
no synthetic pretraining	0.247	0.279	0.344	-0.001	-0.006	-0.419	-0.316
no cavity input	0.259	0.286	0.349	-0.086	-0.028	-0.423	-0.332
scalar sites ($K=0$)	0.215	0.252	0.347	-0.026	-0.003	-0.359	-0.302
Transformer, fine-tuned	0.087	0.121	0.236	-0.163	-0.052	-0.485	-0.381
Residual MLP, fine-tuned	-	-	-	0.025	0.110	-0.246	-0.147

On 16 sensors it is no better than the closed form, and the lifted cavity network beats it by +0.144 [+0.127, +0.161] nats. **Support size.** With 16 calibration rows the transformer degrades sharply, while the lifted cavity network keeps a +0.146 [+0.073, +0.219] nat gain over FA-Gaussian; with 96 rows it remains the best method. **Stronger nonlinearity** (0.8 instead of 0.4) is the one shift where the transformer competes. It ties the lifted cavity network with two missing sensors (difference +0.000 [-0.011, +0.012]) and is better by 0.041 nats when all sensors are observed: under strong response nonlinearity alone, Gaussian re-linearization offers no advantage over a generic transformer. **Linear sensors.** When the true sensors are linear, FA-Gaussian is nearly the right model and is best; the learned re-linearization then costs 0.022 nats with two missing sensors, and the transformer 0.076.

6.4 What the lifting buys

Ablations on F1 (Table 8, Appendix F) isolate the components. **Lifting** is the largest factor: scalar sites ($K=0$) with otherwise identical machinery and training lose 0.134 [+0.120, +0.148] nats with two missing sensors and end up worse than the untrained FA-Gaussian anchor, which is Proposition 1 in practice. **Cavity conditioning** adds 0.026 [+0.022, +0.029] nats on top of lifted static sites. A second nuisance dimension ($K=2$) changes nothing, as expected for one-factor noise. The scalar-cavity network of the earlier design, retrained with the same 60,000-task recipe, remains far behind FA-Gaussian (Appendix D).

6.5 Real sensor networks

Beijing multi-site air quality [36]. The task is to predict $\log \text{PM}_{2.5}$ at each of 12 monitoring stations from the other 11, using hourly data with natural missingness. Each episode is one station and one week, with 48 labeled support hours and 48 query hours. The test period (2016-01 to 2017-02, 718 episodes) was never examined before protocol v2 was committed. Learned models are fine-tuned on episodes from 2013–2015 with one fixed recipe; the protocol’s analysis is cluster-robust over weeks.

With 11 sensors and 48 rows, the joint-covariance closed forms overfit, and evidence-tuned BLR is the strongest closed form. The fine-tuned lifted cavity network improves on it by 0.194 [+0.131, +0.257] nats under natural missingness, and by 0.137 [+0.104, +0.170] with six further sensors removed. The fine-tuned transformer is better still. The lifted cavity network’s gain over it is $-0.118 [-0.145, -0.091]$ nats under natural missingness and $-0.069 [-0.080, -0.058]$ with six sensors removed, so endpoint E4b fails. Before fine-tuning the order is reversed: zero-shot, the lifted cavity network is ahead by 0.127 [+0.057, +0.197] nats with six sensors removed. Fine-tuning improves the transformer by more than it improves the lifted network (Table 3). Synthetic pretraining helps: the same network fine-tuned from initialization is 0.041 nats worse [+0.033, +0.050]. Without fine-tuning, the synthetically trained model transfers poorly (Table 3), reflecting the gap between the synthetic family and real stations.

UCI Air Quality [2]. Five metal-oxide sensors on one device predict the reference analyzers’ $\log \text{CO}$ and $\log \text{NO}_2$, in weekly episodes after October 2004. These weeks were examined in protocol v1, so this is secondary evidence. With two simulated missing sensors, the fine-tuned lifted cavity network improves on FA-Gaussian by +0.107 [+0.015, +0.200] nats for CO and +0.181 [+0.076, +0.287] for NO_2 . The fine-tuned transformer is again better: the lifted network’s gain over it is $-0.060 [-0.106, -0.014]$ (CO) and $-0.044 [-0.074, -0.014]$ (NO_2). On CO , the cavity input stops helping after fine-tuning.

Why the transformer wins on real data. The lifted network underfits rather than overfits: on the Beijing fine-tuning episodes, the transformer’s final training NLL is 0.19 nats lower, about the size of the test gap. Post-hoc variants do not close the gap on Beijing (Table 10; natural missingness, transformer 0.087): a second nuisance dimension scores 0.205 and five times more fine-tuning steps 0.169, against 0.207 for the same seed with the standard recipe. Nor is the edge that of a flexible per-episode regression: a Gaussian process fitted to each episode’s calibration rows is the worst method on Air Quality (3.364 nats for CO with all sensors observed). These point to the Gaussian site family itself. Gaussian sites with per-sensor re-linearization are a strong prior for synthetic factor noise, but real stations share structure they cannot express, and a generic model learns it from a few thousand real episodes: fine-tuning improves the transformer by 0.87 nats on Beijing and the lifted network by 0.62.

6.6 Lifted sites as a transformer output layer (protocol v3)

Protocol v2 left a split: structure wins on synthetic tasks and in extrapolation, while the transformer wins on real networks after fine-tuning. Protocol v3 tests whether the two combine. The *lifted cavity transformer* (LCT) keeps the transformer’s backbone and training recipe unchanged and replaces only its output head. A zero-initialized linear map turns each observed query cell’s final token into a correction $(\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j)$ of that sensor’s lifted site, linearized at the FA-Gaussian posterior mean, and the prediction is the y -marginal of the product of observed sites. Missing sensors therefore contribute no site, the readings of missing sensors never enter, and at initialization the LCT is the FA-Gaussian closed form of its anchor. The hypothesis was formed after seeing the v2 real-data results, so v3 uses only new data: two fresh synthetic panels (G1 with five sensors, G3 with sixteen) and two Beijing pollutants that no model had been scored on (NO_2 and CO , same stations and periods). Its four endpoints were committed before any of these existed (Table 5).

Results are pending.

Table 4: Protocol v3 (new panels and targets). Mean test NLL. G1: 256 fresh five-sensor tasks by number of missing query sensors k ; G3: 128 fresh 16-sensor tasks, $k = 2$. Beijing: log NO₂ and log CO at each station from the other 11 stations, 2016–2017, natural missingness and six further sensors removed; learned models fine-tuned with the recipe of protocol v2 (the all-mask transformer was not fine-tuned). Best entry per column in bold.

Method	G1, $P=5$				G3 $P=16$	Beijing NO ₂		Beijing CO	
	$k=0$	$k=1$	$k=2$	$k=3$		natural	+6	natural	+6
FA-Gaussian	−0.189	−0.054	0.129	0.396	−0.806	0.684	0.506	–	–
Bayesian linear regression	0.013	0.022	0.153	0.401	0.063	0.275	0.365	–	–
Lifted cavity network, 8k	–	–	–	–	–	–	–	–	–
Transformer, 203k	–	–	–	–	–	–	–	–	–
Transformer, all masks, 203k	–	–	–	–	–	–	–	–	–
Lifted cavity transformer, 203k	–	–	–	–	–	–	–	–	–

Table 5: Every pre-registered endpoint of protocols v2 and v3, each committed before its panels or targets existed (protocol v1: Appendix D). Gain in nats of NLL, positive when the first-named model is better; paired 95% intervals, cluster-robust over weeks on Beijing. Pass: gain ≥ 0.01 and lower bound > 0 , except E7b (lower bound > -0.02).

ID	Data	Comparison	Gain [95% CI]	Result
E1	F1, $k = 2$	lifted cavity network vs NL-FA	+0.054 [+0.043, +0.065]	pass
E2	F1, $k = 2$	lifted cavity network vs transformer	+0.033 [+0.027, +0.040]	pass
E3a	F3 (16 sensors), $k = 2$	lifted cavity network vs FA-Gaussian	+0.148 [+0.115, +0.181]	pass
E3b	F3 (16 sensors), $k = 2$	lifted cavity network vs transformer	+0.144 [+0.127, +0.161]	pass
E4a	Beijing PM _{2.5}	lifted cavity network (FT) vs BLR	+0.194 [+0.131, +0.257]	pass
E4b	Beijing PM _{2.5}	lifted cavity network (FT) vs transformer (FT)	−0.118 [−0.145, −0.091]	fail
E5	G1, $k = 2$	LCT vs transformer	pending	–
E6	G3 (16 sensors), $k = 2$	LCT vs transformer	pending	–
E7a	Beijing NO ₂	LCT (FT) vs lifted cavity network (FT)	pending	–
E7b	Beijing NO ₂	LCT (FT) vs transformer (FT), non-inferiority	pending	–

6.7 Pre-registered endpoints and calibration

Table 5 lists every pre-registered endpoint of protocols v2 and v3. Five of the six v2 endpoints pass; the failure, E4b, is the real-data comparison with the fine-tuned transformer. All five endpoints of protocol v1 passed (Appendix D). The v3 endpoints are pending.

Every learned model is close to nominal 90% coverage on F1 (lifted cavity network 0.901, transformer 0.902), while the closed forms under-cover (0.860–0.873). On Beijing both fine-tuned networks cover 0.914, against 0.807–0.861 for the closed forms. CRPS and RMSE rank the methods as NLL does (Table 9 in Appendix F).

7 Limitations

Scope. The exact and Bayes-optimal references exist only for our generator (one-factor shared noise, a sinusoidal response nonlinearity, Gaussian targets), and the shift panels vary width, nonlinearity and support size within it. On the Air Quality device, per-sensor dropout is simulated because its own outages remove all sensors at once. In Beijing, natural missingness is real but rare (1.6% of query readings), and larger deletions are simulated. The gas-sensor-drift benchmark could not separate methods because heavy outliers dominate Gaussian likelihoods (Appendix D). **Model class.** Gaussian sites over a low-dimensional latent cannot represent multimodal posteriors, and on real networks they underfit (§6.5). We used $K = 1$ throughout. **Baselines.** Each transformer was trained once, with a CPU budget (203k parameters, 30,000 steps). Pretrained tabular foundation models (TabPFN v2, TabICL) could not be run in our sandbox; we provide a script but report no numbers for them.

8 Conclusion

In-context sensor fusion must handle missing sensors and shared noise at the same time. A product of experts over the target handles the first by construction but provably mishandles the second. Lifting each sensor’s site to a target-plus-noise latent restores exactness for factor-structured noise

under every missingness pattern. A learned cavity-conditioned refinement on top of a closed-form factor-analysis anchor turns this into a small model that:

- outperforms strong closed forms and a 25-times-larger TabPFN-v2-style transformer trained on the same task distribution;
- extrapolates to unseen missingness patterns and to more sensors than it was trained on;
- fine-tunes to real sensor networks.

Natural next steps are vector targets, robust site likelihoods for heavy-tailed sensors, mixtures of sites for multimodal posteriors, and evaluation on further deployments, including comparisons with pretrained tabular foundation models.

References

- [1] T. W. Anderson and Herman Rubin. Statistical inference in factor analysis. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, volume 5, pages 111–150, 1956.
- [2] Saverio De Vito, Ettore Massera, Marco Piga, Luca Martinotto, and Girolamo Di Francia. On field calibration of an electronic nose for benzene estimation in an urban pollution monitoring scenario. *Sensors and Actuators B: Chemical*, 129(2):750–757, 2008.
- [3] Marc Peter Deisenroth and Shakir Mohamed. Expectation propagation in Gaussian process dynamical systems. In *Advances in Neural Information Processing Systems*, volume 25, 2012.
- [4] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1):1–38, 1977.
- [5] Ruifeng Dong, Ming Wang, Ning Liu, Tong Guo, Jiayi Kang, Xiaojing Shen, and Yao Mao. DIFNet: Decentralized information filtering fusion neural network with unknown correlation in sensor measurement noises. *arXiv preprint arXiv:2508.18854*, 2025.
- [6] S. M. Ali Eslami, Daniel Tarlow, Pushmeet Kohli, and John Winn. Just-in-time learning for fast and flexible inference. In *Advances in Neural Information Processing Systems*, volume 27, 2014.
- [7] Ángel F. García-Fernández, Lennart Svensson, Mark R. Morelande, and Simo Särkkä. Posterior linearization filter: Principles and implementation using sigma points. *IEEE Transactions on Signal Processing*, 63(20):5561–5573, 2015.
- [8] Marta Garnelo, Dan Rosenbaum, Christopher Maddison, Tiago Ramalho, David Saxton, Murray Shanahan, Yee Whye Teh, Danilo Rezende, and S. M. Ali Eslami. Conditional neural processes. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1704–1713, 2018.
- [9] Samuel J. Gershman and Noah D. Goodman. Amortized inference in probabilistic reasoning. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 36, 2014.
- [10] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [11] Karol Gregor and Yann LeCun. Learning fast approximations of sparse coding. In *Proceedings of the 27th International Conference on Machine Learning*, pages 399–406, 2010.
- [12] Nicolas Heess, Daniel Tarlow, and John Winn. Learning to pass expectation propagation messages. In *Advances in Neural Information Processing Systems*, volume 26, 2013.
- [13] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002.
- [14] Noah Hollmann, Samuel Müller, Katharina Eggersperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations*, 2023.

- [15] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326, 2025.
- [16] Wittawat Jitkrittum, Arthur Gretton, Nicolas Heess, S. M. Ali Eslami, Balaji Lakshminarayanan, Dino Sejdinovic, and Zoltán Szabó. Kernel-based just-in-time learning for passing expectation propagation messages. In *Proceedings of the 31st Conference on Uncertainty in Artificial Intelligence (UAI)*, 2015.
- [17] Julie Josse, Nicolas Prost, Erwan Scornet, and Gaël Varoquaux. On the consistency of supervised learning with missing values. *arXiv preprint arXiv:1902.06931*, 2019.
- [18] Simon J. Julier and Jeffrey K. Uhlmann. A non-divergent estimation algorithm in the presence of unknown correlations. In *Proceedings of the American Control Conference*, volume 4, pages 2369–2373, 1997.
- [19] Hyunjik Kim, Andriy Mnih, Jonathan Schwarz, Marta Garnelo, Ali Eslami, Dan Rosenbaum, Oriol Vinyals, and Yee Whye Teh. Attentive neural processes. In *International Conference on Learning Representations*, 2019.
- [20] Marine Le Morvan, Julie Josse, Thomas Moreau, Erwan Scornet, and Gaël Varoquaux. NeuMiss networks: differentiable programming for supervised learning with missing values. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- [21] Roderick J. A. Little and Donald B. Rubin. *Statistical Analysis with Missing Data*. Wiley, 3 edition, 2019.
- [22] Mengmeng Ma, Jian Ren, Long Zhao, Sergey Tulyakov, Cathy Wu, and Xi Peng. SMIL: Multimodal learning with severely missing modality. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 2302–2310, 2021.
- [23] David J. C. MacKay. Bayesian interpolation. *Neural Computation*, 4(3):415–447, 1992.
- [24] Thomas P. Minka. Expectation propagation for approximate Bayesian inference. In *Proceedings of the 17th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 362–369, 2001.
- [25] Vishal Monga, Yuelong Li, and Yonina C. Eldar. Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Processing Magazine*, 38(2):18–44, 2021.
- [26] Samuel Müller, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. Transformers can do Bayesian inference. In *International Conference on Learning Representations*, 2022.
- [27] Radford M. Neal. MCMC using Hamiltonian dynamics. In Steve Brooks, Andrew Gelman, Galin Jones, and Xiao-Li Meng, editors, *Handbook of Markov Chain Monte Carlo*. Chapman and Hall/CRC, 2011.
- [28] Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A tabular foundation model for in-context learning on large data. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pages 50817–50847, 2025.
- [29] Guy Revach, Nir Shlezinger, Xiaoyong Ni, Adrià López Escoriza, Ruud J. G. van Sloun, and Yonina C. Eldar. KalmanNet: Neural network aided Kalman filtering for partially known dynamics. *IEEE Transactions on Signal Processing*, 70:1532–1547, 2022.
- [30] Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976.
- [31] Donald B. Rubin and Dorothy T. Thayer. EM algorithms for ML factor analysis. *Psychometrika*, 47(1):69–76, 1982.
- [32] Alexander Vergara, Shankar Vembu, Tuba Ayhan, Margaret A. Ryan, Margie L. Homer, and Ramón Huerta. Chemical gas sensor drift compensation using classifier ensembles. *Sensors and Actuators B: Chemical*, 166–167:320–329, 2012.

- 395 [33] M. Wang, H. Liu, H. Tang, M. Zhang, and X. Shen. IFNet: Data-driven multisensor estimate
396 fusion with unknown correlation in sensor measurement noises. *Information Fusion*, 115, 2025.
- 397 [34] Mike Wu and Noah Goodman. Multimodal generative models for scalable weakly-supervised
398 learning. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- 399 [35] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov,
400 and Alexander Smola. Deep sets. In *Advances in Neural Information Processing Systems*,
401 volume 30, 2017.
- 402 [36] Shuyi Zhang, Bin Guo, Anlan Dong, Jing He, Ziping Xu, and Song Xi Chen. Cautionary
403 tales on air-quality improvement in Beijing. *Proceedings of the Royal Society A*, 473(2205):
404 20170457, 2017.

A Proofs

Proposition 1. Under the model, $p(x_j | y) = \mathcal{N}(x_j; a_j y, c_j)$ with $c_j = d_j^2 + \sigma_j^2$.

If $q(y | x_j) = p(y | x_j)$ for all x_j , then $p(y) t_j(y; x_j) \propto p(y) p(x_j | y)$ as functions of y . Hence $t_j(y; x_j) = \kappa_j(x_j) p(x_j | y)$, which is Gaussian in y with precision $a_j^2/c_j > 0$. For a pair, q is therefore the naive-Bayes posterior $p(y) p(x_j | y) p(x_k | y)$. Its posterior mean is linear in (x_j, x_k) with coefficients

$$(a_j/c_j, a_k/c_k) / \pi_q, \quad \pi_q = 1 + a_j^2/c_j + a_k^2/c_k,$$

so the coefficient of x_j has the sign of a_j .

The Bayes posterior uses $C = \text{Cov}(x_j, x_k | y)$, with off-diagonal entry $d_j d_k$. Its mean coefficients are $(C^{-1}a)^\top / \pi$, with $\pi = 1 + a^\top C^{-1}a$. Both posteriors are Gaussian, so they coincide for all (x_j, x_k) if and only if $C^{-1}a = \text{diag}(C)^{-1}a$. Equivalently $a = C \text{diag}(C)^{-1}a$, whose off-diagonal components are $d_j d_k a_k/c_k = 0$ and $d_j d_k a_j/c_j = 0$. With $a_j a_k \neq 0$ this holds if and only if $d_j d_k = 0$.

For the sign flip, take $a = (1, 1)$, $d = (1, 0.5)$ and $\sigma = (0.05, 0.05)$. Then the Bayes coefficient of x_1 is $+0.499$ alone and -0.959 in the pair. $\square$

Proposition 2. Given $z = (y, h)$, the readings are independent: $x_j | z \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$. Hence

$$p(z | x_S) \propto p(z) \prod_{j \in S} \mathcal{N}(x_j; w_j^\top z + c_j, \psi_j).$$

In natural parameters each factor contributes $w_j w_j^\top / \psi_j$ to the precision and $w_j(x_j - c_j) / \psi_j$ to the shift, which is (2) with $o_j = x_j - c_j$. The prior is Gaussian, so the product is the Gaussian (3). This holds for every S because the factors of unobserved sensors are absent from both sides. Marginalizing h gives $p(y | x_S)$. $\square$

Proposition 3. *Step 1: the product is a latent-factor model.* By the computation in the proof of Proposition 2 read backwards, a product of sites (2) with $o_j = x_j - c_j$ is the exact posterior over z of the model $M_q: z \sim \mathcal{N}(\mu_z, \Sigma_z)$ and, independently given z , $x_j \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$. Its y -marginal is M_q 's conditional $p_q(y | x_S)$, and M_q 's sensor covariance is $\Sigma_q = W \Sigma_z W^\top + \text{diag}(\psi)$ with $W \in \mathbb{R}^{P \times (1+K')}$, so $\Sigma_q - \text{diag}(\psi)$ has rank at most $1 + K'$.

Step 2: conditionals on $|S| \leq 2$ identify the covariance. Write $V = \text{Var}(y)$, $c = \text{Cov}(x, y)$ and $\Sigma = \text{Cov}(x)$. The case $S = \emptyset$ gives V . For $S = \{j\}$ the conditional has slope $\beta_j = c_j / \Sigma_{jj}$ and variance $r_j = V - c_j \beta_j$, and $\beta_j \neq 0$ because $c_j = a_j \neq 0$. Hence $c_j = (V - r_j) / \beta_j$ and $\Sigma_{jj} = c_j / \beta_j$. For $S = \{j, k\}$ the slope vector $b \neq 0$ solves $\Sigma_{\{jk\}} b = (c_j, c_k)$, and whichever entry of b is non-zero determines Σ_{jk} from one row. Since M_q has the same conditionals, the same formulas apply to it, so $\Sigma_q = \Sigma = LL^\top + \text{diag}(\sigma^2)$.

Step 3: fewer factors cannot reproduce Σ . Let $r = 1 + K$, so $LL^\top$ has rank r . Suppose $\Sigma = G + \Psi'$ with Ψ' diagonal and $\text{rank}(G) \leq r$. Fix a row i and two disjoint row sets I_1, I_2 not containing i with L_{I_1}, L_{I_2} nonsingular. In the $(r+1) \times (r+1)$ submatrix of Σ with rows $\{i\} \cup I_1$ and columns $\{i\} \cup I_2$, every entry except (i, i) lies off the diagonal, so it is shared by G and $LL^\top$. Both G and $LL^\top$ have rank at most r , so the corresponding minors vanish. Expanding along entry (i, i) , its cofactor is $\det(L_{I_1} L_{I_2}^\top) \neq 0$, so $G_{ii} = (LL^\top)_{ii}$. Hence $G = LL^\top$, which has rank r [1]. Applying this to $G = W \Sigma_z W^\top$ from Step 1 gives $1 + K' \geq r = 1 + K$.

For almost every L all $r \times r$ submatrices are nonsingular, and $P \geq 2r + 1 = 2K + 3$ leaves two disjoint sets of r rows after deleting any row. $\square$

Proposition 4.

- (i) Sites of missing sensors are multiplied by the mask before entering (3), and the network's per-sensor inputs never include another sensor's raw reading, only cavity statistics built from observed sites.

- 448 (ii) Every operation is applied per sensor with shared weights. EM and principal factors are
 449 equivariant to permutations of the sensor coordinates, and sums over sensors are permutation-
 450 invariant. No input encodes a sensor’s index or the total number of sensors, except the
 451 fraction of observed sensors, which is permutation invariant.
- 452 (iii) With a zero last layer, $\delta^{\text{off}} = \delta^{\text{slope}} = \delta^D = s = 0$. Then $w_j = (A_j, D_j)$, ψ_j is unchanged
 453 and $o_j = \tilde{x}_j - A_j(\bar{y} - \mu_y) + A_j\bar{y} = \tilde{x}_j + A_j\mu_y$. These are the initialization sites, so every
 454 round reproduces them and the output is the anchor’s FA-Gaussian closed form. The zero
 455 point lies in the parameter set, so the infimum of the training risk is at most FA-Gaussian’s.
 456 $\square$

457 B Architecture and training details

458 Lifted cavity network ($K = 1$, $R = 3$, 8,004 parameters).

- 459 • **Support encoder.** $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^{32} \rightarrow \mathbb{R}^{32}$ with tanh, applied to each observed calibration
 460 entry $(\tilde{y}_i, \tilde{x}_{ij}, \tilde{r}_{ij})$. The inputs are the standardized target, the centred reading, and the
 461 anchor residual scaled by $\sqrt{\psi_j}$. The encoder averages over observed rows and then applies
 462 $\rho : \mathbb{R}^{33} \rightarrow \mathbb{R}^{16}$, where the extra input is the observation rate.
- 463 • **Site network.** $f_\theta : \mathbb{R}^{28} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^4$ with tanh and a zero-initialized last layer.
- 464 • **Anchor.** EM uses 8 iterations with a 10^{-4} ridge. Principal factors use 40 iterations, with
 465 uniquenesses floored at 10^{-3} . Everything is computed in double precision from the support
 466 only.

Table 6: Compute. Training on one CPU core (core-hours); inference (ms/task) is single-threaded time per task for 480 queries (ten masks $\times$ 48 rows) on F1. Anchors are precomputed in batch and excluded; the closed-form rows include their unbatched NumPy fits.

Model	Parameters	Training steps	Core-hours	ms/task
Lifted cavity network	8,004	20,000	0.22	7.2
Residual MLP on FA anchor	15,490	20,000	0.20	1.2
Transformer (TabPFN-v2 style)	202,562	30,000	3.11	42.9
Lifted cavity transformer	—	—	—	—
FA-Gaussian (closed form)	—	—	—	38.3
NL-FA (closed form)	—	—	—	34.4

467 Training recipe (all synthetic-trained models except the transformer).

- 468 • Pool of 60,000 tasks (seed 5150).
- 469 • Minibatches of 128 tasks $\times$ 8 queries.
- 470 • AdamW with learning rate 1.8×10^{-3} , cosine decay to 10% and weight decay 10^{-4} .
- 471 • Gradient clipping at 5 and 20,000 steps.
- 472 • Seeds 1–3.

473 No hyperparameter search was run for any model.

474 Controls.

- 475 • **Residual MLP on the same anchor** (15,490 parameters). It receives the FA-Gaussian
 476 prediction, the same per-sensor support encodings (concatenated in sensor order), the
 477 readings and mask, and a 55-dimensional support summary. It outputs a residual correction
 478 to the FA-Gaussian mean and log-variance, and is defined only for five sensors.
- 479 • **Lifted sites without cavity input.** The same sites with the cavity statistics removed.
- 480 • **1-D sites** ($K = 0$). The same machinery, with sites over y only.
- 481 • **Repository cavity network.** The three-round scalar-cavity architecture of Appendix D,
 482 retrained with the shared recipe.

483 **Transformer baseline (202,562 parameters).** The transformer follows TabPFN v2 [15]:

- 484 • every cell is a token, embedded by a linear map of its value or by a learned missing-value
485 token;
- 486 • sensors receive random per-task feature embeddings, so the model is width-agnostic and
487 permutation equivariant in distribution;
- 488 • the target column has its own embedding, and query targets use a learned token;
- 489 • each of 4 blocks alternates attention across the columns of a row with attention across rows
490 within a column (all rows attend to the support rows only), followed by an MLP ($d = 64$, 4
491 heads, hidden width 128, pre-LayerNorm);
- 492 • targets are standardized by the support mean and standard deviation, and the output is a
493 Gaussian mean and log-variance.

494 It was trained on fresh tasks at every step (16 tasks $\times$ 16 queries, AdamW with learning rate 10^{-3} ,
495 1,000 warm-up steps, cosine decay, clipping at 1, 30,000 steps), with the same query masks as the
496 other models. A second copy (*all masks*) saw query masks with 0–4 missing sensors, drawn uniformly.
497 Evaluation uses a fixed seed for the random feature embeddings.

498 **Lifted cavity transformer (protocol v3, 202,822 parameters).** The backbone is the transformer
499 above, unchanged. Its output head is replaced by a zero-initialized linear map from each query cell’s
500 final token to four numbers ($\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j$), used exactly as in (4) with the linearization point $\bar{y}$
501 set to the anchor’s closed-form posterior mean (one round, $K = 1$, gradients not propagated through
502 $\bar{y}$). Readings of missing sensors are replaced by the missing-value token before the backbone and
503 contribute no site, so they never enter the prediction (a unit test checks both this and the identity with
504 the anchor’s closed form at initialization). Anchors are computed on the fly for each training task.
505 Training, masks, seed and evaluation are those of the transformer.

506 C Baselines and references

507 Support-only closed forms.

- 508 • **EM-Gaussian:** the conditional of y given x_S under the EM joint Gaussian (40 iterations).
- 509 • **FA-Gaussian:** the closed form of §4, with EM run for 40 iterations. The networks’ anchors
510 use 8 iterations and no variance inflation, so the reported FA-Gaussian is a slightly stronger
511 baseline than the untrained networks (on F1 with two missing sensors, 0.177 against 0.187).
- 512 • **NL-FA:** per-sensor cubic response curves in the standardized target. The intercept and slope
513 are unpenalized; the curvature terms are ridge-penalized ($\lambda = 100$) and clamped to the
514 support’s target range for linear extrapolation. The residual covariance has a one-factor
515 structure. Exact inference is done on a 401-point grid. The degree and λ were tuned on
516 the development panel only: free degree selection by leave-one-out produced catastrophic
517 extrapolation.
- 518 • **BLR:** evidence-maximized Bayesian linear regression [23] on complete-case rows, with
519 predictive variance including parameter uncertainty.
- 520 • **Ridge:** complete-case ridge with leave-one-out residual variance, and the original mean-
521 imputed ridge.
- 522 • **Gaussian process (post hoc):** for each episode and observed-sensor set, an ARD squared-
523 exponential kernel plus white noise on the complete-case calibration rows, with hyperpa-
524 rameters by type-II maximum likelihood (one restart).

525 EM-Gaussian, FA-Gaussian and ridge inflate their predictive variance by $n/(n - |S| - 1)$.

526 **Bayes-optimal ceiling.** The ceiling is HMC [27] over the 25 (or $5P$) task parameters. It uses
527 the true priors, transformed to unconstrained coordinates. The shared nuisance is marginalized
528 analytically through a rank-one-plus-diagonal covariance, using only the observed entries of each
529 calibration row. The sign of each a_j is fixed by the support correlation, which is unambiguous
530 because $|a_j| \geq 0.6$.

The sampler runs 4 chains per task, each started from a MAP fit. Warm-up has two phases: step-size adaptation (target acceptance 0.75) and diagonal mass-matrix estimation from 150 iterations, followed by 150 more adaptation iterations. We keep 50 draws per chain, thinned by 4, with 12 leapfrog steps.

The predictive density of a query uses the samples exactly, including the query’s own reweighting: $p(y | x_S, \mathcal{D}) = \sum_m p(y, x_S | \theta_m) / \sum_m p(x_S | \theta_m)$, with $p(x_S | \theta_m)$ by trapezoid quadrature on 301 points. On every panel where it was run, R-hat on identifiable parameters was below 1.1 for at least 99.9% of parameters. Run with the true parameters, the predictive formula reproduces the exact oracle to within 2×10^{-3} nats (a unit test).

D Protocols and audit trail

How the method arose. The lifted cavity network replaced an earlier design that failed. That design was a three-round EP-style network whose sites live on y alone and are updated from the scalar cavity of y . It was evaluated under pre-registered gates on the same synthetic family. Its first version collapsed to the prior during training. A repaired version, with sites initialized from support moments, trained successfully but was beaten by a ridge regression. On that panel the oracle, the ridge control and the best repaired seed scored:

Method	NLL, two sensors missing
Oracle	−0.046
Ridge control	0.396
Repaired scalar-cavity network (best seed)	0.403

Computing the oracle exposed three problems. First, the control was weak: it mean-imputed missing calibration entries, and a support-only EM or factor-analysis Gaussian scores 0.168 or 0.156. Second, scalar sites are structurally limited (§3). Third, the learners were starved: 512 reused tasks and 3,000 updates. Retraining the scalar-cavity architecture with our recipe (60,000 tasks, 20,000 steps) improves it by only about 0.05 nats, and on F1 it remains 0.159 nats worse than the untrained FA-Gaussian (§6.4).

Protocol v1. After development on panels that are now retired, five endpoints were committed to version control before their panels were generated:

- fresh synthetic tasks, against FA-Gaussian and against the residual MLP;
- 8-sensor transfer;
- stronger nonlinearity;
- a new real target (NO₂ on the Air Quality device).

All five passed (Table 7). Each endpoint was evaluated once, and no model was retrained after the commit.

Table 7: Protocol v1 endpoints (two missing sensors; positive gain favours the lifted cavity network; pass requires gain ≥ 0.01 and a positive lower bound).

Endpoint	Comparison	Gain [95% CI]	Result
E1	Fresh tasks, vs FA-Gaussian	+0.074 [+0.064, +0.084]	pass
E2	Fresh tasks, vs residual MLP	+0.016 [+0.012, +0.020]	pass
E3	8 sensors, vs FA-Gaussian	+0.115 [+0.097, +0.132]	pass
E4	Nonlinearity 0.8, vs residual MLP	+0.028 [+0.020, +0.036]	pass
R1	Air Quality NO ₂ , 23 later weeks, vs EM-Gaussian	+0.112 [+0.050, +0.173]	pass

Protocol v2. Protocol v2 was likewise committed before any of its panels or test splits existed. It adds:

- the transformer baseline, NL-FA, BLR and the HMC ceiling;
- wider panels (8 and 16 sensors), and shifts in nonlinearity and support size;

Table 8: Ablations on F1 (mean test NLL). Same training recipe for every learned row.

Variant	$k=0$	$k=1$	$k=2$	$k=3$
Lifted cavity, $K = 1$ (ours)	-0.265	-0.112	0.097	0.395
Lifted cavity, $K = 2$	-0.267	-0.112	0.098	0.397
Lifted sites, no cavity input	-0.221	-0.074	0.123	0.399
Scalar sites ($K = 0$), same machinery	-0.070	0.061	0.232	0.473
Scalar-cavity network (retrained)	0.055	0.178	0.336	0.555
FA-Gaussian (untrained)	-0.152	-0.011	0.177	0.439

Table 9: Calibration and point accuracy. Coverage of the central 90% predictive interval (target 0.90), continuous ranked probability score (CRPS, lower is better) and RMSE. F1 with two missing sensors; Beijing test with natural missingness (fine-tuned learned models).

Method	F1, $k = 2$			Beijing, natural		
	Cov. 90	CRPS	RMSE	Cov. 90	CRPS	RMSE
FA-Gaussian	0.873	0.168	0.316	0.824	0.189	0.379
EM-Gaussian	0.860	0.170	0.318	0.807	0.207	0.423
BLR	0.866	0.174	0.327	0.861	0.182	0.374
Lifted cavity (ours)	0.901	0.160	0.305	0.914	0.171	0.357
Transformer	0.902	0.165	0.314	0.914	0.158	0.341
Residual MLP	0.902	0.162	0.309	—	—	—

- the Beijing benchmark, whose test period had never been scored.

NL-FA’s settings and BLR’s choice as the strongest real-data closed form were fixed on development data (Appendix C). The gas-sensor-drift dataset [32] was examined during development and then excluded, before any test split was built. Its heavy-tailed sensor outliers made the mean NLL of every Gaussian method dominated by a few episodes, with mean and median differing by more than 100 nats, so it cannot separate methods. One deviation was recorded before any Beijing score existed: some query hours have no other station observed, and complete-case BLR cannot fit zero features, so for an empty sensor set it returns the support mean with variance $s^2(1 + 1/n)$, like the other closed forms’ prior predictive.

Protocol v3. After protocol v2’s real-data endpoint failed, protocol v3 was committed before any of its panels or targets existed. It fixes one model (the lifted cavity transformer of §6.6, trained once with the transformer’s recipe and seed and evaluated at its final checkpoint), two new synthetic panels (G1, G3), two Beijing pollutants never scored by any model (NO₂, primary; CO, descriptive), the fine-tuning recipe of protocol v2, and four endpoints (Table 5). The lifted cavity transformer’s scores on the v2 panels are descriptive only (Appendix F).

Post-hoc analyses. Everything not named in a protocol is post hoc and labeled as such: the per-episode Gaussian process, the fine-tuning diagnostics of §6.5, and the exploratory runs of Appendix F.

E Broader impact

Calibrating low-cost environmental sensors widens access to air-quality monitoring. Overconfident predictions could mislead exposure decisions, which is why we report coverage alongside likelihood. We recommend validating calibration on each new deployment before relying on predicted intervals.

F Additional results

Table 10: Post-hoc fine-tuning variants on Beijing PM_{2.5} (test period of protocol v2; mean test NLL). Neither more nuisance dimensions nor longer fine-tuning closes the gap to the fine-tuned transformer.

Fine-tuned model	natural	+3	+6
Lifted cavity network, standard recipe (seed 1)	0.207	0.237	0.304
with $K = 2$	0.205	0.239	0.300
with $5\times$ fine-tuning steps	0.169	0.201	0.276
Transformer, standard recipe	0.087	0.121	0.236

Table 11: Panel F2 (8 sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	−0.807	−0.712	−0.598	−0.459
FA-Gaussian	−0.462	−0.387	−0.296	−0.183
EM-Gaussian	0.032	−0.133	−0.160	−0.120
NL-FA	−0.485	−0.414	−0.327	−0.215
BLR	0.187	0.398	0.615	0.028
Ridge (complete case)	0.064	−0.003	−0.033	−0.015
Lifted cavity (ours)	−0.615	−0.530	−0.427	−0.302
Lifted cavity, $K = 2$	−0.618	−0.531	−0.427	−0.302
Lifted sites, no cavity	−0.551	−0.469	−0.369	−0.250
Scalar sites ($K = 0$)	−0.295	−0.228	−0.148	−0.057
Transformer	−0.568	−0.474	−0.362	−0.229
Transformer, all masks	−0.545	−0.452	−0.343	−0.212

Table 12: Panel F3 (16 sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	−1.223	−1.184	−1.142	−1.096
FA-Gaussian	−0.864	−0.832	−0.797	−0.758
EM-Gaussian	6.300	4.851	2.977	1.553
NL-FA	−0.910	−0.876	−0.841	−0.803
BLR	0.095	0.078	0.058	0.054
Ridge (complete case)	0.093	0.089	0.081	0.086
Lifted cavity (ours)	−1.018	−0.982	−0.945	−0.902
Lifted cavity, $K = 2$	−1.017	−0.981	−0.944	−0.901
Lifted sites, no cavity	−0.952	−0.916	−0.880	−0.836
Scalar sites ($K = 0$)	−0.658	−0.627	−0.591	−0.554
Transformer	−0.886	−0.846	−0.801	−0.747
Transformer, all masks	−0.869	−0.832	−0.790	−0.738

Table 13: Panel F4 (nonlinearity 0.8; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	−0.513	−0.331	−0.086	0.261
FA-Gaussian	0.050	0.170	0.334	0.568
EM-Gaussian	0.050	0.145	0.311	0.559
NL-FA	−0.032	0.089	0.258	0.507
BLR	0.208	0.188	0.331	0.568
Ridge (complete case)	0.138	0.208	0.338	0.571
Lifted cavity (ours)	−0.185	−0.026	0.191	0.498
Lifted cavity, $K = 2$	−0.200	−0.035	0.186	0.497
Lifted sites, no cavity	−0.068	0.075	0.261	0.519
Scalar sites ($K = 0$)	−0.013	0.130	0.314	0.570
Transformer	−0.226	−0.043	0.191	0.503
Transformer, all masks	−0.221	−0.038	0.190	0.486
Residual MLP	−0.107	0.039	0.235	0.508
Scalar-cavity network	0.133	0.256	0.416	0.635

Table 14: Panel F5 (linear sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	-0.455	-0.288	-0.066	0.244
FA-Gaussian	-0.362	-0.203	0.009	0.305
EM-Gaussian	-0.244	-0.142	0.034	0.308
NL-FA	-0.337	-0.178	0.030	0.318
BLR	-0.135	-0.085	0.061	0.322
Ridge (complete case)	-0.169	-0.092	0.062	0.323
Lifted cavity (ours)	-0.323	-0.172	0.031	0.318
Lifted cavity, $K = 2$	-0.320	-0.170	0.033	0.318
Lifted sites, no cavity	-0.343	-0.189	0.018	0.309
Scalar sites ($K = 0$)	-0.085	0.031	0.186	0.415
Transformer	-0.278	-0.122	0.085	0.371
Transformer, all masks	-0.281	-0.123	0.080	0.353
Residual MLP	-0.304	-0.161	0.036	0.319
Scalar-cavity network	0.052	0.161	0.302	0.505

Table 15: Panel F6 (16 support rows; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	-0.446	-0.280	-0.060	0.252
FA-Gaussian	0.256	0.371	0.510	0.686
EM-Gaussian	24.900	4.682	1.140	0.739
NL-FA	0.314	0.388	0.496	0.661
BLR	0.898	0.857	3.690	0.887
Ridge (complete case)	0.317	0.391	0.500	0.624
Lifted cavity (ours)	0.073	0.195	0.363	0.604
Lifted cavity, $K = 2$	0.156	0.250	0.392	0.614
Lifted sites, no cavity	0.117	0.234	0.389	0.604
Scalar sites ($K = 0$)	0.204	0.293	0.428	0.644
Transformer	0.253	0.371	0.523	0.731
Transformer, all masks	0.256	0.359	0.483	0.653
Residual MLP	0.094	0.211	0.371	0.594
Scalar-cavity network	0.345	0.439	0.565	0.750

Table 16: Panel F7 (96 support rows; 128 tasks): mean test NLL by number of missing sensors.

Method	$k=0$	$k=1$	$k=2$	$k=3$
Oracle	-0.429	-0.264	-0.043	0.269
FA-Gaussian	-0.232	-0.082	0.114	0.385
EM-Gaussian	-0.237	-0.086	0.112	0.385
NL-FA	-0.276	-0.127	0.073	0.353
BLR	-0.178	-0.056	0.125	0.390
Ridge (complete case)	-0.163	-0.047	0.129	0.391
Lifted cavity (ours)	-0.324	-0.163	0.052	0.353
Lifted cavity, $K = 2$	-0.320	-0.159	0.056	0.357
Lifted sites, no cavity	-0.275	-0.123	0.078	0.359
Scalar sites ($K = 0$)	-0.097	0.034	0.205	0.443
Transformer	-0.295	-0.133	0.081	0.379
Transformer, all masks	-0.297	-0.134	0.080	0.367
Residual MLP	-0.287	-0.131	0.074	0.360
Scalar-cavity network	0.047	0.161	0.309	0.516

Table 17: Air Quality CO: mean test NLL for every number of simulated missing sensors (23 later weeks).

Method	$k=0$	$k=1$	$k=2$	$k=3$
EM-Gaussian	0.074	0.063	0.085	0.146
FA-Gaussian	0.052	0.086	0.116	0.154
NL-FA	0.144	0.120	0.091	0.124
Bayesian linear regression	1.037	0.433	0.216	0.202
Ridge (complete case)	0.824	0.337	0.208	0.210
GP, complete case [†]	3.364	1.056	0.503	0.146
Lifted cavity, zero-shot	0.044	0.064	0.129	0.160
Transformer, zero-shot	−0.017	−0.006	0.081	0.154
Lifted cavity, fine-tuned (ours)	−0.040	−0.031	0.008	0.077
no synthetic pretraining	−0.001	−0.014	−0.006	0.077
no cavity input	−0.086	−0.065	−0.028	0.038
scalar sites ($K = 0$)	−0.026	−0.021	−0.003	0.075
Transformer, fine-tuned	−0.163	−0.149	−0.052	0.036
Residual MLP, fine-tuned	0.025	0.061	0.110	0.160

Table 18: Air Quality NO₂: mean test NLL for every number of simulated missing sensors (23 later weeks).

Method	$k=0$	$k=1$	$k=2$	$k=3$
EM-Gaussian	−0.237	−0.174	−0.175	−0.168
FA-Gaussian	−0.196	−0.173	−0.155	−0.162
NL-FA	−0.090	−0.102	−0.142	−0.169
Bayesian linear regression	−0.055	−0.214	−0.178	−0.160
Ridge (complete case)	−0.200	−0.230	−0.176	−0.145
Lifted cavity, zero-shot	−0.350	−0.295	−0.233	−0.176
Transformer, zero-shot	−0.360	−0.310	−0.231	−0.136
Lifted cavity, fine-tuned (ours)	−0.412	−0.366	−0.337	−0.273
no synthetic pretraining	−0.419	−0.365	−0.316	−0.248
no cavity input	−0.423	−0.375	−0.332	−0.275
scalar sites ($K = 0$)	−0.359	−0.337	−0.302	−0.231
Transformer, fine-tuned	−0.485	−0.429	−0.381	−0.294
Residual MLP, fine-tuned	−0.246	−0.198	−0.147	−0.106

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